

TECHNICAL SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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S.1. DECISION ACCURACY, TRANSPORT, AND STOPPING: PROOFS

This section gives the complete arguments for the new economic error chain. The inherited verification, Bellman-bridge, and diffusion proofs are retained in the preceding sections. The constants below pertain to the new stored finite-law experiment unless a more general hypothesis is stated.

S.1.1. *Proof of Proposition 1*

First suppose an optimal action a^* exists. The approximate maximizing property implies

$$r_\theta(s, a^*) + \hat{S}_0(T_\theta(s, a^*)) - r_\theta(s, \hat{a}) - \hat{S}_0(T_\theta(s, \hat{a})) \leq \delta_\theta(s).$$

Substituting $S_\theta = \hat{S}_0 - e + \Delta_\theta$ gives

$$\begin{aligned} Q_\theta(s, a^*) - Q_\theta(s, \hat{a}) &\leq \delta_\theta(s) + e(T_\theta(s, \hat{a})) - e(T_\theta(s, a^*)) \\ &\quad + \Delta_\theta(T_\theta(s, a^*)) - \Delta_\theta(T_\theta(s, \hat{a})). \end{aligned}$$

Each difference is at most its action-set oscillation. For an unattained finite supremum, take an ϵ -maximizer and let ϵ decrease to zero. This proves (13) without a differentiability assumption.

For finite actions let $v_j = e(T_\theta(s, a_j)) - \bar{e}_\pi(s)$. In the weighted inner product $\langle x, y \rangle_\pi = \sum_j \pi_j x_j y_j$, the representer of $v_i - v_j$ has entries $1/\pi_i$ at i , $-1/\pi_j$ at j , and zero elsewhere. Its squared norm is $1/\pi_i + 1/\pi_j$. Cauchy–Schwarz gives

$$(v_i - v_j)^2 \leq (1/\pi_i + 1/\pi_j) \sum_k \pi_k v_k^2.$$

Maximizing over the finite pair set proves $\text{osc } e \leq \sqrt{C_\pi R_\pi}$. The transport difference is at most $2b_\theta$, proving (14). For random fitted objects the inequality is pathwise.

With deterministic C_π , for example, $\mathbb{E}\sqrt{C_\pi R_\pi} \leq \sqrt{C_\pi \mathbb{E}R_\pi}$ whenever the expectation is finite. Repetition of the same random seed does not create a new draw for this expectation. Finally, if every nonoptimal action loses at least g and the displayed bound is below g , a nonoptimal implemented action would contradict the bound. $\square$

S.1.2. Proof of Theorem 2

Couple both economies by the same innovation sequence and the same initial postdecision state. If $\|X_k^\theta - X_k^0\| \leq d_k$, then the triangle inequality and (15) give

$$\|X_{k+1}^\theta - X_{k+1}^0\| \leq \epsilon_{\theta,k} + L_k d_k = d_{k+1}.$$

Induction starts at $d_0 = 0$. At each reward date, $|g_{\theta,k}(X_k^\theta) - g_{0,k}(X_k^0)| \leq \eta_{\theta,k} + H_k d_k$. Multiply by the nonnegative weights, sum including the terminal reward, and take expectations. The bound is uniform because each assumed bound is uniform on the invariant domain. No interchange of a supremum over future controls with a fixed-policy expectation is used.

For recursive evaluations write $D_k = \|V_{\theta,k} - V_{0,k}\|_\infty$. Add and subtract $\mathcal{G}_{\theta,k}(V_{0,k+1})$ to obtain $D_k \leq \nu_{\theta,k} + \beta_k D_{k+1}$. Backward substitution from $D_K \leq b_K$ gives (17). Both arguments require that every coupled state and every recursively evaluated utility belong to the domains on which the bounds were verified. The theorem makes no extension past these domains. If a domain is localized, an exit contribution must be bounded separately before using the result as a global guarantee. $\square$

S.1.3. Proof of Proposition 4

Strong concavity gives, for any feasible b ,

$$Q(b) - Q(a) \leq Q'(a)(b - a) - \kappa(b - a)^2/2.$$

In the interior, maximize this upper bound over all real $b - a$ to obtain $|Q'(a)|^2/(2\kappa)$. At the lower endpoint restrict the maximization to $b - a \geq 0$. Its value is

$$\frac{(\max\{Q'(a), 0\})^2}{2\kappa}.$$

At the upper endpoint the analogous residual is $\max\{-Q'(a), 0\}$. Enlarging a bounded feasible half-interval to the half-line is conservative. This proves (20).

An outward derivative interval bounds the applicable residual, and a positive lower curvature bound enlarges the resulting nonnegative fraction. Thus each task certificate holds deterministically at every attempted stage. A first successful average certificate implies the average economic target; previous failures do not affect validity. If checks are stochastic, let B_j denote the event that attempted check j falsely certifies its candidate. The conditional hypothesis gives $\Pr(B_j) \leq \mathbb{E}\alpha_j$; an unattempted check contributes zero. The union bound yields $\Pr(\cup_j B_j) \leq \mathbb{E}\sum_j \alpha_j \leq \alpha$. This proof requires conditional validity for the data actually used after past decisions. A fixed-time interval repeatedly queried without such a property does not meet the hypothesis.

The work identity follows by summing the disjoint measured operations through the stop, including unsuccessful fits and tests. Under the stated exact linear cost model, subtraction gives $W_N(q) - W_R(q) = F_N - F_R - q(c_R - c_N)$, establishing the strict inequality in (21). Integer query counts use that strict inequality; a tie at an integer threshold is not a strict saving. Measured costs in the experiment are used directly and are not assumed to satisfy a linear model. $\square$

S.1.4. Proof of Corollary 3

Condition on the feature map and design, which are independent of the fitting noises. Least squares gives $\hat{\beta} - \beta = G_n^{-1}\Phi^\top \xi$. For a fixed queried argument x , the prediction error is a linear combination of the independent noises. Its sub-Gaussian variance parameter is at most

$$\sigma^2 \|\Phi G_n^{-1} \phi(x)\|^2 = \sigma^2 \phi(x)^\top G_n^{-1} \phi(x) \leq \sigma^2 \ell / n.$$

The two-sided sub-Gaussian bound and a union bound over the K distinct arguments therefore give

$$\Pr\left(\max_x |\hat{S}(x) - S(x)| > \sigma \sqrt{\frac{2\ell}{n} \log \frac{4K}{\alpha}}\right) \leq \alpha/2.$$

Uniformly on that event, the oscillation in every feasible finite action menu is at most twice the displayed bound. Proposition 1 with zero transport and (18) gives loss at most ε at every query. Full column rank and the leverage bound are separate hypotheses and must actually hold at the selected integer n . They are not consequences of the scalar sample-size inequality.

For the direct method, each prediction average has sub-Gaussian variance parameter at most σ^2/t . Dependence between predictions caused by common innovations across different arguments does not affect the union bound; independence is required only for the observations within an average, or an equivalent sub-Gaussian bound for that average. The same calculation with ℓ/n replaced by $1/t$ proves the second guarantee. A union bound over the two methods gives joint coverage $1 - \alpha$, without assuming independence between them. Reusing a cached prediction at exactly the same continuation identity and argument incurs no additional simulation count. This is why K , rather than the number of repeated requests, enters the work comparison.

Subtract the complete specified costs to obtain (19). The fixed term $F(n)$ includes dictionary construction, all labels, the least-squares solve and its validation; it is not just a neural update clock. Common selection and verification work cancels only under the explicit equal-cost hypothesis. Otherwise its difference remains. These are sufficient budgets and actual costs for the declared procedures, not statistical lower bounds on arbitrary algorithms. If $|\mathbb{E}\hat{S}(x) - S(x)| \leq B$ uniformly under a misspecified dictionary, the prediction bound acquires B ; if future continuation changes by at most b , the decision bound acquires a further $2b$. Thus the same argument uses margin $\varepsilon - \delta - 2B - 2b > 0$, provided the stated bias bound is verified. A pointwise approximation error of an arbitrary basis expansion is not automatically a bound on the bias of its fitted least-squares projection. $\square$

S.1.5. Uniform curvature in the stored capital economy

Let $M = (1 - \mu)I + \mu\mathbf{1}\mathbf{1}^\top/d$. It is row stochastic and has infinity operator norm one. For $f(y) = \lambda \tanh(My)$, the inequalities $|\tanh'| \leq 1$ and $|\tanh''| \leq 2$ imply,

along a scalar path with $\|y'\|_\infty \leq P$ and $\|y''\|_\infty \leq D$,

$$\|(f(y))''\|_\infty \leq \lambda(D + 2P^2). \quad (\text{S.1})$$

The postdecision map has first derivative $-h\mathbf{1}$ and second derivative zero, so initialize $P = h$, $D = 0$. Adding the first innovation does not change these derivatives. For each of the three future transitions set

$$B_k = \lambda(D + 2P^2), \quad D \leftarrow D + hB_k, \quad P \leftarrow (1 + h\lambda)P.$$

The order matters: B_k uses the preceding P, D . The second derivative of the future production reward is at most $W_k B_k$, since every stored W_k is positive. All future terms involving the fixed withdrawal are constant with respect to the current action.

For a vector v write $\text{range}(v) = \max_i v_i - \min_i v_i$. At the first shocked postdecision state a valid range bound for each stored path is

$$R_0 = \text{range}(y) + 2h\lambda + \text{range}(z_0).$$

Every subsequent transition satisfies $R_k = R_{k-1} + 2h\lambda + \text{range}(z_k)$ as an upper bound. The scalar drift and common withdrawal cancel from a coordinate range. For the terminal payoff $g(y) = -\gamma \text{Var}_d(y)/2$,

$$(g(y))'' = -\gamma \text{Var}_d(y') - \gamma \overline{(y - \bar{y}\mathbf{1})y''} \leq \gamma R_3 D_3.$$

The first term is nonpositive; dropping it is an upper bound, not a sign reversal. Combining these inequalities gives a uniform upper bound on the continuation curvature,

$$S_0(x(y, a))'' \leq M(y) := \sum_{k=1}^3 W_k B_k + \gamma \mathbb{E}_{16}[R_3 D_3].$$

Since the current reward has second derivative $-A_0(w/a^2 + \eta)$, a valid lower bound for true strong concavity on the whole interval is

$$\kappa(y, \theta) = A_0\{w/\bar{a}^2 + \eta\} - M(y). \quad (\text{S.2})$$

The verifier evaluates this expression with outward arithmetic and requires a strictly positive lower endpoint for every task. A nonpositive result would be a verification

failure, not a reason to delete that task. The published records retain every returned lower endpoint.

The true derivative is enclosed separately rather than inferred from the surrogate. For each stored path initialize $p_0 = -h\mathbf{1}$. At a future state,

$$f'(y) = \lambda\{1 - \tanh^2(My)\}Mp, \quad p \leftarrow p + hf'(y).$$

Accumulate $W_k \overline{f'(y)}$ and the terminal derivative $-\gamma \overline{(y - \bar{y}\mathbf{1})p}$. Average over all sixteen atoms and add the current derivative $A_0(w/a - \eta a - \tau) - W_0$. Every operation is evaluated on enclosing intervals. Dependency between occurrences of the same variable may widen an interval, but cannot invalidate the enclosure.

S.1.6. Arithmetic enclosure and its implementation

The stored binary64 coefficients and innovation increments define the exact finite economy. Floating-point evaluation of a Gaussian generator is not being asserted to be exact sampling from a continuous Gaussian economy. The generator is used to construct and freeze a finite array; the subsequent scientific statement concerns that array with equal masses.

The arithmetic module encloses each binary operation by the adjacent representable numbers toward minus and plus infinity. It checks finite values, IEEE round-to-nearest behavior and gradual underflow. Products and quotients take extrema over endpoint combinations, with divisors bounded away from zero. Squaring includes zero when the argument interval contains zero. Balanced reductions avoid interpreting a non-enclosing ordinary mean as an interval sum. Induction on an expression's operation graph proves inclusion for arithmetic expressions from enclosed inputs.

The nonlinear enclosure does not assume that a library's hyperbolic tangent is correctly rounded. For $0 \leq t \leq 1/2$, the exponential is enclosed by its degree-18 Taylor polynomial and the nonnegative tail bound

$$0 \leq e^t - \sum_{j=0}^{18} t^j / j! \leq \frac{t^{19} / 19!}{1 - t/20}.$$

All terms are enclosed using the same arithmetic. Exact binary scaling (checked on every replayed argument) reduces the argument to this interval; repeated squaring and reciprocation recover positive and negative arguments. The identity $\tanh x = (e^{2x} - 1)/(e^{2x} + 1)$ and monotonicity provide endpoint enclosures. For the auxiliary logarithm check, scaling to $[1, 2)$ gives $z = (x - 1)/(x + 1) \in [0, 1/3)$ and

$$\log x = 2 \sum_{j=0}^{31} \frac{z^{2j+1}}{2j+1} + R, \quad 0 \leq R \leq \frac{2z^{65}}{65(1-z^2)}.$$

The stored enclosing endpoints for $\log 2$ is independently checked with exact rational series bounds. Synthetic high-precision derivative tests check the implemented recursion on separate small-dimensional cases. These tests complement the analytical enclosure argument; they are not substitutes for it or additional economic observations.

S.1.7. Charge monotonicity and the economic tolerance

For an interior maximizing action, differentiate $Q'_\theta(y, a^*(\tau)) = 0$. Because the charge enters the derivative as $-A_0\tau$, the implicit-function identity is $Q'' \partial a^*/\partial \tau - A_0 = 0$, proving (29). For the constrained case take $\tau_2 > \tau_1$ and respective optimal actions a_2, a_1 . Adding their optimality inequalities leaves $A_0(\tau_2 - \tau_1)(a_2 - a_1) \leq 0$, hence $a_2 \leq a_1$. Uniqueness follows from strict concavity.

An external grant in the utility account satisfies $A_0 w \log(1 + g) = U$ exactly when $g = \exp(U/(A_0 w)) - 1$. This is a monotone normalization of a certified payoff loss. The published grant table uses ordinary binary64 evaluation and is explicitly descriptive; the primary rigorous statement is the outward payoff-loss bound. No claim is made that increasing actual endogenous consumption by this amount leaves capital, future rewards, or the reference policy unchanged.

S.2. FROM CURRENT DECISIONS TO POLICY ACCURACY

A continuation certificate is useful for a dynamic economy only if the certified object is the future of the policy under assessment. This section states the conditions under which the local decision accounts compose. They apply to additive and order-

preserving recursive evaluation, while keeping the utility domains and admissible deviations of the economic model.

At dates $t = 0, \dots, T - 1$, let $\mathcal{D}_t$ be an invariant state domain and $A_t(s)$ the full admissible current action set. Write

$$(T_t v)(s) = \sup_{a \in A_t(s)} G_t(s, a, v), \quad (T_t^\pi v)(s) = G_t(s, \pi_t(s), v).$$

Assume the dynamic programming principle on these domains, measurable admissible policies, and finite values. Each G_t is order preserving in its continuation argument and is sup-norm Lipschitz, uniformly in (s, a) , with constant $\beta_t \geq 0$ on a common invariant value domain. Let $V_T^* = V_T^\pi$ and define $V_t^* = T_t V_{t+1}^*$ and $V_t^\pi = T_t^\pi V_{t+1}^\pi$. The relevant local advantage is

$$g_t^\pi(s) = T_t V_{t+1}^\pi(s) - T_t^\pi V_{t+1}^\pi(s). \quad (\text{S.3})$$

It evaluates all current deviations against the *same implemented future policy*; it is neither a training residual nor an advantage computed against an unrelated anchor continuation.

THEOREM S.1—Composition of economic certificates: *Under the preceding assumptions, suppose that on a common event*

$$0 \leq g_t^\pi(s) \leq \varepsilon_t \quad \text{for every } s \in \mathcal{D}_t, \quad t = 0, \dots, T - 1. \quad (\text{S.4})$$

Then, on that event, at every initial state,

$$0 \leq V_0^* - V_0^\pi \leq \sum_{t=0}^{T-1} w_t \varepsilon_t, \quad w_t = \prod_{j=0}^{t-1} \beta_j. \quad (\text{S.5})$$

For additive Markov evaluation $G_t(s, a, v) = r_t(s, a) + \beta_t P_t^a v(s)$, a state-dependent refinement is $V_t^ - V_t^\pi \leq B_t$, where*

$$B_T = 0, \quad B_t(s) = u_t(s) + \beta_t \sup_{a \in A_t(s)} P_t^a B_{t+1}(s) \quad (\text{S.6})$$

whenever $u_t(s) \geq g_t^\pi(s)$ and the displayed quantities are finite.

Section S.3 proves the result. Centered continuation-error and transport bounds from Proposition 1 can supply (S.4) when established uniformly on the required state and action domains. Alternatively, an interval certificate for the true one-step objective can supply it directly. In either case, the continuation must be V_{t+1}^π , or its discrepancy must be charged. The verification may be deterministic. With random certificates, all domain and date bounds must hold on one event, for example by a prespecified summable conditional error allocation. Selecting a stopping time does not remove this requirement.

This theorem identifies a positive route from reusable policy evaluation to full policy accuracy. It also identifies precisely what the computational evidence below supplies. A mean certificate on a finite state catalogue establishes the declared current-decision experiment; it does not establish (S.4) outside that catalogue. Sampling only states visited by π is insufficient because a profitable deviation may reach other states. The state envelope in (S.6) deliberately takes a supremum over admissible transitions, not only the transitions of the fitted policy.

S.2.0.0.1. Relation to the continuous economy and strategic applications. Let Π_h be a specified implementable policy class in the original economy. If its class-approximation gap is at most η_{class} , and $|J(\pi) - J_h(\pi)| \leq \eta_{\text{value}}$ uniformly for $\pi \in \Pi_h$, then a policy satisfying (S.5) in the approximating model has

$$\sup_{\pi} J(\pi) - J(\hat{\pi}) \leq \eta_{\text{class}} + 2\eta_{\text{value}} + \sum_t w_t \varepsilon_t + \eta_{\text{impl}}, \quad (\text{S.7})$$

where η_{impl} bounds any additional payoff loss in implementation. The class and value-transfer allowances are separate mathematical inputs, not numbers inferred from the finite experiments. For recursive utility, order and Lipschitz bounds are imposed on the actual common utility domain; actionwise centering is applied to the evaluated action payoff, not through an arbitrary nonlinear aggregator. For a dynamic game, apply the same account to each player's full deviation problem against the same opponents' profile. Bounds ϵ_i for all players imply an ϵ_i -best-response profile under the specified equilibrium concept. A single local deviation or a single player's calculation does not supply those premises. The original informa-

tion, boundary, transversality, and deviation requirements remain in the Economic Applications companion.

S.2.1. Allocating computation across dates

The policy target suggests how to distribute a verified approximation budget. Suppose the local account at date t is $b_t + e_t$, where b_t contains fixed optimization, representation, and transport allowances. Suppose further that a declared procedure attains approximation allowance $e_t > 0$ using $n_t \geq A_t/e_t^2$ fitting observations at charged marginal cost $c_t > 0$. Here $A_t > 0$ includes its confidence allocation and design constants. The full-rank and coverage conditions underlying such a rate must hold at the selected budgets; Corollary 3 is one finite-query example, not a license to assume uniform coverage of an infinite state space.

PROPOSITION S.2—Allocation for a policy target: Assume $w_t > 0$ and let $E = \varepsilon - \sum_t w_t b_t > 0$, $a_t = c_t A_t$, and $H = \sum_t a_t^{1/3} w_t^{2/3}$. Among positive allowances satisfying $\sum_t w_t e_t \leq E$, the minimum continuous fitting bill is

$$\min \sum_t \frac{a_t}{e_t^2} = \frac{H^3}{E^2}, \quad e_t = \frac{E}{H} \left(\frac{a_t}{w_t} \right)^{1/3}. \quad (\text{S.8})$$

Rounding each observation count up adds at most $\sum_t c_t$ to this bill. All fixed construction, reusable-cache, query, and verification costs must be added. The statement minimizes the displayed sufficient-cost account, not the work of all possible numerical methods.

The propagation and allocation arguments use standard dynamic-programming stability and convex optimization. Their role here is to connect the paper’s centered, transported, and true-objective certificates to a specified dynamic economic target. They do not replace approximation or coverage assumptions. Finite-time fitted-value analysis, including the distinction between sampling norms and policy loss, provides an important antecedent (Munos and Szepesvári, 2008). The empirical contribution below remains a measured comparison of complete procedures, rather than a claim of a new principle of optimality.

S.3. PROOFS FOR POLICY COMPOSITION AND ECONOMIC CONTRASTS

PROOF OF THEOREM S.1: Backward induction and order preservation give $V_t^* \geq V_t^\pi$. Suppose $0 \leq V_{t+1}^* - V_{t+1}^\pi \leq b$ on $\mathcal{D}_{t+1}$. The elementary supremum inequality and the uniform Lipschitz condition give

$$0 \leq T_t V_{t+1}^* - T_t V_{t+1}^\pi \leq \beta_t b.$$

Adding $T_t V_{t+1}^\pi - V_t^\pi = g_t^\pi$ shows that the sup-norm error is at most $\varepsilon_t + \beta_t b$. Iteration from the common terminal value proves (S.5), including the case of a zero Lipschitz constant. No assumption that π itself attains the optimal action is used. For the state envelope, suppose by induction that $V_{t+1}^* - V_{t+1}^\pi \leq B_{t+1}$. Then

$$\begin{aligned} V_t^*(s) - V_t^\pi(s) &\leq \sup_a \{r_t(s, a) + \beta_t P_t^a V_{t+1}^\pi(s) + \beta_t P_t^a B_{t+1}(s)\} - V_t^\pi(s) \\ &\leq g_t^\pi(s) + \beta_t \sup_a P_t^a B_{t+1}(s) \leq B_t(s). \end{aligned}$$

Invariance ensures that every expectation is taken on a domain on which the induction applies. These are pathwise deductions on the common certificate event, so no additional probability statement is introduced by the proof. *Q.E.D.*

S.3.0.0.1. *Why catalogue averages cannot be substituted.* Consider a two-date economy with zero first-date rewards. A deviation at the initial state can reach a state with a unit second-date gain that the implemented policy never visits. All local advantages at the implemented policy's visited states can be zero while the initial optimality gap equals one. Thus a bound only on that occupancy distribution cannot be substituted for (S.4). The same example shows why changing the supremum in (S.6) to the transition of π is generally invalid. Weighted sampling-norm results require their own distribution-shift or coverage assumptions. We make no such inference from the finite query banks.

PROOF OF THE BRIDGE (S.7): For the returned approximating policy $\pi_h \in \Pi_h$, write

$$\sup_{\pi} J(\pi) - J(\hat{\pi}) \leq \eta_{\text{class}} + \sup_{\pi \in \Pi_h} J(\pi) - J(\pi_h) + \eta_{\text{impl}}$$

$$\leq \eta_{\text{class}} + 2\eta_{\text{value}} + \sup_{\pi \in \Pi_h} J_h(\pi) - J_h(\pi_h) + \eta_{\text{impl}}.$$

Apply Theorem S.1 to the last difference with the specified policy class and action information. If an enlarged approximating action class is used, its optimum is an upper bound and the inequality continues to hold. The conclusion for fixed opponents follows by taking this same supremum over each player's admissible deviations. A simultaneous profile of these inequalities is precisely the stated approximate best-response condition; no existence or fixed-point claim is inferred. *Q.E.D.*

PROOF OF PROPOSITION S.2: For any positive feasible allowances, Hölder's inequality implies

$$H = \sum_t \left(\frac{a_t}{e_t^2} \right)^{1/3} (w_t e_t)^{2/3} \leq \left(\sum_t \frac{a_t}{e_t^2} \right)^{1/3} \left(\sum_t w_t e_t \right)^{2/3}.$$

Hence the objective is at least H^3/E^2 . The displayed allowances satisfy $\sum_t w_t e_t = E$ and attain equality. Since $0 \leq \lceil A_t/e_t^2 \rceil - A_t/e_t^2 < 1$, their integer observation cost exceeds the continuous bill by less than $\sum_t c_t$. At any date whose statistical design requires a minimum sample size, that requirement must also hold at the rounded budget; otherwise the design must be charged and the constrained allocation recomputed. Dates with zero propagation weight are omitted from this positive-weight allocation problem. *Q.E.D.*

S.3.1. Optimal-action intervals and comparative statics

The economic bands in Table XV use more information than a mean value-loss certificate. Let Q be continuously differentiable and κ -strongly concave on $[\ell, u]$, with unique maximizer a^* . For feasible a , let $r(a)$ be the directional residual in Proposition 4. Then

$$|a - a^*| \leq r(a)/\kappa. \quad (\text{S.9})$$

To see this for $a < a^*$, the optimality condition gives $Q'(a^*) \geq 0$ when $a^* = u$, and gives zero at an interior optimum. Strong concavity then implies $Q'(a) \geq \kappa(a^* - a)$. For $a > a^*$, the reverse argument gives $-Q'(a) \geq \kappa(a - a^*)$. At an endpoint the

positive-part residual gives exactly the appropriate one-sided inequality. The case $a = a^*$ is immediate.

Use the interval upper bound $\bar{r}_i$ and the strictly positive curvature lower bound $\underline{\kappa}_i$ for each task. Outward arithmetic and intersection with feasibility give

$$I_i = \left[a_i - \frac{\bar{r}_i}{\underline{\kappa}_i}, a_i + \frac{\bar{r}_i}{\underline{\kappa}_i} \right] \cap [\ell_i, u_i], \quad a_i^* \in I_i.$$

For the same tasks under a changed future θ and the anchor, the mean optimal response belongs to $q^{-1} \sum_i (I_{\theta,i} - I_{0,i})$. No independent-noise assumption is needed: these are deterministic inclusions for fixed finite laws. We use the final NBO actions, not the enumeration actions, to obtain the NBO response bands. The envelope over three fitting streams contains every corresponding inclusion; it does not turn three fixed streams into a population confidence interval. Printed endpoints are rounded outwards. A negative upper endpoint establishes a negative mean optimal response in that declared economy.

S.3.2. Source and arithmetic audit

The R21 audit reads every original R20 service record, verifies its stored SHA256 digest, reestablishes the inherited source identities, regenerates only the fixed task arguments, and recomputes the certificates at the saved actions. It does not retrain any predictor or replace any clock. All 984 attempted certificates and 120 additional zero-charge certificates are compared field for field with their original values. The attempted checks cover 319,062 task actions. The audit also verifies feasibility, stopping order, recorded success flags, every cost decomposition, and the exact binary scaling used in the interval exponential implementation. This is an arithmetic and source audit of the specified finite model, not an empirical assessment of external economic calibration.

S.4. FULL-POLICY VERIFICATION DETAILS

This section supplies the arithmetic and implementation accounts for Section 6. All numerical coefficients are the exact values of their stored binary64 representa-

tions. The stochastic law remains Gaussian with those coefficients; no finite shock catalogue is substituted.

S.4.1. The quadratic local advantage

For a fixed supplied future policy, substitute $U_{t+1}(z) = z'P_{t+1}z + c_{t+1}$ into its one-step cost. The action-dependent terms are

$$u'H_t u + 2\beta u'B_t'P_{t+1}A_t x.$$

Since $H_t \succ 0$, the minimum over the entire space $\mathbb{R}^d$ occurs at $u_t^g = -H_t^{-1}\beta B_t'P_{t+1}A_t x$. Evaluating at $-K_t x$ and subtracting this minimum yields

$$(-K_t x - u_t^g)' H_t (-K_t x - u_t^g) = x' D_t' H_t^{-1} D_t x.$$

There is no approximation in this identity and no reference to the optimal future policy. The possibly loose Frobenius bound in (30) is chosen to make the implementation independent of an uncertified spectral decomposition. A failed bound need not indicate that the actual policy loss exceeds the tolerance.

S.4.2. Outward matrix balls

A matrix ball (C, R) denotes every matrix M with $|M - C| \leq R$ entrywise, where $R \geq 0$. Exact input balls have zero radius. For conformable balls (C, R) and (D, S) , let k be the inner dimension. The exact product is enclosed about the computed product $\text{fl}(CD)$ by the radius

$$\gamma_{2k}|C||D| + |C|S + R|D| + RS + 2k\tau\mathbf{1}, \quad (\text{S.10})$$

with every nonnegative calculation rounded upward. Here $\gamma_n = nu/(1 - nu)$, u is the binary64 unit roundoff, and τ is the smallest positive normal binary64 number. The underflow allowance deliberately exceeds the corresponding subnormal absolute-error allowance. The factor $2k$ permits separate multiplication and addition rounding in classical binary64 products. Fused operations only reduce this particular account. Overflow and nonfinite values are rejected rather than certified.

To calculate a nonnegative product upper bound, the implementation divides the computed product by a downward bound for $1 - \gamma_{2k}$, adds the underflow allowance, and rounds upward. Addition, scalar multiplication, traces, squared Frobenius norms, and maximum absolute row sums have analogous outward bounds. These operations enclose the actual recursion, not merely a rounded residual evaluated without error bars. Positivity of P_t follows from the exact evaluation equation and positive stage costs, not from an unverified numerical eigenvalue. In the executed model the positive diagonal entries of Q_t, R_t give the coercivity lower bounds directly. Covariance validity is checked by symmetry and strict diagonal dominance.

The tests compare matrix operations with rational arithmetic on the exact binary input values, including signed products and cancellation. Separate tests compare policy losses with a separately computed optimal comparator, check the coercive bound at states far outside an ordinary training range, perturb policies within the declared execution class, and verify that the certificate still runs when the Riccati comparator is disabled. These tests complement, rather than replace, the enclosure argument.

S.4.3. A robust action-implementation envelope

Suppose the implemented action is $-K_t x + e_t$, with $\|e_t\|_2 \leq e\|x\|_2$, possibly depending on the past and current state. Put $\bar{P}_T = Q_T$ and $\bar{c}_T = 0$. For $F_t = A_t - B_t K_t$, set

$$h_t = 2\|R_t K_t\|_F e + \|R_t\|_\infty e^2 + \beta\{2\|\bar{P}_{t+1} F_t\|_F \|B_t\|_F e + \|\bar{P}_{t+1}\|_\infty \|B_t\|_F^2 e^2\}, \quad (\text{S.11})$$

$$\bar{P}_t = Q_t + K_t' R_t K_t + \beta F_t' \bar{P}_{t+1} F_t + h_t I,$$

$$\bar{c}_t = \beta\{\bar{c}_{t+1} + \text{tr}(\bar{P}_{t+1} \Sigma)\}. \quad (\text{S.12})$$

All displayed coefficients are replaced by their outward upper enclosures in the numerical calculation.

PROPOSITION S.3—Implementation allowance: *For every execution satisfying the relative action-error bound, its remaining cost is at most $x'\bar{P}_tx + \bar{c}_t$. On the initial ball $\|x\|_2^2 \leq b$, its excess over the ideal stored policy is at most*

$$\varepsilon_{\text{impl}} = \max\{0, b\|\bar{P}_0 - P_0\|_\infty + \bar{c}_0 - c_0\}. \quad (\text{S.13})$$

PROOF: Expanding the current action cost around $-K_tx$ bounds its increase by $\{2\|R_tK_t\|_F e + \|R_t\|_\infty e^2\}\|x\|_2^2$. Expanding the next-state quadratic envelope around F_tx bounds its increase by the second line of (S.11), times $\|x\|_2^2$. The mean-zero innovation contributes its trace term and has no cross term with a nonanticipating action. Backward induction proves the value envelope. Subtract the exact stored-policy quadratic value, use symmetry and the maximum-row-sum bound, and maximize over the initial ball. This proves (S.13) even when the execution errors are history dependent. *Q.E.D.*

The error budget is explicit. Certification of arbitrary machine executions without an error contract would be a different claim. The experiment neither removes this term nor infers it from agreement on sampled states.

S.4.4. Reoptimized comparative statics

Let $K^{(r)}$ be a certified complete policy in regime r , and let its reported total loss bound on the initial ball be b_r . At a fixed initial state, the first-action objective with the *optimal* future is a strictly convex quadratic with Hessian at least $2R_0$. The cost difference between using the candidate's first action and the optimal first action, both followed by the optimal future, is no larger than the candidate's full policy loss. Thus, with $r_0 \leq \lambda_{\min}(R_0)$,

$$\|u_0^{(r)} - u_0^{*,(r)}\|_2 \leq \sqrt{b_r/r_0}. \quad (\text{S.14})$$

At $x_0 = -\mathbf{1}$, define mean investment $m_r = d^{-1}\mathbf{1}'K_0^{(r)}\mathbf{1}$. Its distance from the optimal mean is at most $\sqrt{b_r/(r_0d)}$. The reported radius also includes the relative-execution and summation allowances. Subtracting the interval for the anchor from that for the changed regime gives a valid optimum-response interval. It is an enclosure in a

specified model, not a confidence interval for an estimated economic population.
Strong competitors are evaluated by the same construction.

S.4.5. Trainable continuation factors

For the square-activation critic, write $L(W; P) = \frac{1}{4} \|W'W - dP\|_F^2$, with symmetric own-policy target P . Differentiation gives $\nabla_W L = W(W'W - dP)$. If $X \sim N(0, I)$, the mean squared error between the gradients of $\|WX\|^2/d$ and $X'PX$ equals $4\|W'W - dP\|_F^2/d^2$. The target is obtained by evaluating the current candidate policy; it is not the Riccati value matrix. The experiment trains all hidden weights for a fixed number of gradient updates and then solves the actor's full vector-action problem. A conventional quadratic receives the identical population target and is allowed its exact minimizer. This matched-information comparison isolates the cost of the neural factorization. No advantage over conventional quadratics is built into the target oracle.

S.5. RECURSIVE-RISK AND EXECUTION ENVELOPES

LEMMA S.4—Gaussian continuation perturbations: Let $P \succeq L \succeq 0$, $\|P - L\|_2 \leq d_0$, and $\gamma = 1 - 2\theta\|\Sigma\|_2\|P\|_2 > 0$. Then

$$0 \preceq \Psi_\theta(P) - \Psi_\theta(L) \preceq \frac{d_0}{\gamma^2} I, \quad 0 \leq h_\theta(P) - h_\theta(L) \leq \frac{d_0 \operatorname{tr} \Sigma}{\gamma}.$$

The same inequalities hold at $\theta = 0$ by continuity.

PROOF: Gaussian integration after completing the square gives (35). Its derivatives in a symmetric direction E are

$$D\Psi_\theta(P)[E] = (I - 2\theta P\Sigma)^{-1}E(I - 2\theta\Sigma P)^{-1},$$

and

$$Dh_\theta(P)[E] = \operatorname{tr}\{\Sigma^{1/2}(I - 2\theta\Sigma^{1/2}P\Sigma^{1/2})^{-1}\Sigma^{1/2}E\}.$$

Along the segment from L to P , the inverse norms in the first display are at most γ^{-1} by the Neumann series. The symmetric inverse in the second display is bounded

by $\gamma^{-1}I$. Integrating the derivatives, using $0 \preceq P - L \preceq d_0 I$, proves the claims. This proof also applies to singular positive semidefinite Σ . Q.E.D.

PROOF OF THEOREM 8: Set $L_t(x) = x'(P_t - \delta_t I)x + c_t - \kappa_t$. The stated checks ensure that every continuation coefficient used in this construction is positive semidefinite and inside the Gaussian moment domain. Terminally $L_T = U_T$. At date t , write $P = P_{t+1}$ and $L = P - \delta_{t+1}I$. Let $S(P)$ denote the coefficient of the minimized current-action objective with continuation coefficient P . Completion of squares gives

$$P_t - S(P) = D_t' H_t^{-1} D_t \preceq \zeta_t^2 I / r_t.$$

Let K^L minimize the objective using $\Psi_\theta(L)$, and $H^L = R_t + \beta B_t' \Psi_\theta(L) B_t$. The exact identity

$$H^L(K_t - K^L) = D_t + \beta B_t' \{\Psi_\theta(P) - \Psi_\theta(L)\} F_t$$

and Lemma S.4 imply $\|A_t - B_t K^L\|_2 \leq \bar{f}_t$. Evaluating the larger-continuation objective at K^L therefore gives

$$0 \preceq S(P) - S(L) \preceq \beta \bar{f}_t^2 w_t I.$$

Consequently $P_t - \delta_t I \preceq S(L)$. Own-policy constants satisfy $c_t = \beta\{c_{t+1} + h_\theta(P_{t+1})\}$. The constant part of (36) and the lemma give

$$c_t - \kappa_t \leq \beta\{c_{t+1} - \kappa_{t+1} + h_\theta(L)\}.$$

Together these are precisely the Bellman-subsolution inequality $L_t \leq \inf_u \{\ell_t + \beta \mathcal{R}_\theta L_{t+1}\}$. Order preservation and backward induction imply $L_0 \leq J_0^*$, including comparison with arbitrary history-adapted controls of finite recursive cost. The feasible stored policy gives $J_0^* \leq U_0$. Subtracting proves (37). Q.E.D.

PROPOSITION S.5—Execution errors under recursive preferences: *Suppose the implemented action differs from $-K_t x$ by an adapted error e_t satisfying $\|e_t\|_2 \leq \varepsilon_{\text{ex}} \|x\|_2$ at every date. Let $k_t \geq \|K_t\|_2$ and $s_t \geq \|R_t\|_2$. Starting with $\delta_T^e = \kappa_T^e = 0$, define*

$$\gamma_t^e = 1 - 2\theta\sigma(p_{t+1} + \delta_{t+1}^e), \quad \nu_t = (p_{t+1} + \delta_{t+1}^e) / \gamma_t^e,$$

$$a_t^e = 2\varepsilon_{\text{ex}}k_t s_t + \varepsilon_{\text{ex}}^2 s_t + \beta v_t(2\varepsilon_{\text{ex}}f_t b_t + \varepsilon_{\text{ex}}^2 b_t^2),$$

$$\delta_t^e = \beta f_t^2 \delta_{t+1}^e / (\gamma_t^e)^2 + a_t^e, \quad \kappa_t^e = \beta \{\kappa_{t+1}^e + \tau \delta_{t+1}^e / \gamma_t^e\}.$$

If every $\gamma_t^e > 0$, the implemented policy's cost is at most $U_0(x) + \delta_0^e \|x\|^2 + \kappa_0^e$. Its excess cost over J^* is thus bounded by $(\delta_0 + \delta_0^e) \|x\|^2 + \kappa_0 + \kappa_0^e$.

PROOF: Assume the next implemented value is bounded by $x'(P_{t+1} + \delta_{t+1}^e I)x + c_{t+1} + \kappa_{t+1}^e$. Apply Lemma S.4 between P_{t+1} and $P_{t+1} + \delta_{t+1}^e I$, using the margin γ_t^e . The unchanged transition contributes at most $\beta f_t^2 \delta_{t+1}^e \|x\|^2 / (\gamma_t^e)^2$. The action perturbation contributes at most $(2\varepsilon_{\text{ex}}k_t s_t + \varepsilon_{\text{ex}}^2 s_t) \|x\|^2$ to current cost and at most $\beta v_t(2\varepsilon_{\text{ex}}f_t b_t + \varepsilon_{\text{ex}}^2 b_t^2) \|x\|^2$ to the continuation quadratic. The log-determinant and constant contributions are bounded by κ_t^e . Backward induction proves the upper envelope. The last assertion uses Theorem 8. The action-error premise is an explicit mathematical contract; it is not inferred merely from using floating-point hardware on unbounded inputs. *Q.E.D.*

S.5.0.0.1. *Verified numerical inverses.* For an interval matrix S , propose a numerical inverse C and bound $r \geq \|I - SC\|_\infty$. If $r < 1$, then

$$\|S^{-1} - C\|_\infty \leq \|C\|_\infty r / (1 - r).$$

This follows from $S^{-1} = C(I - (I - SC))^{-1}$. A matrix ball with that uniform entry radius encloses the inverse. The implementation uses this test for $S = I - 2\theta\Sigma P$, together with the moment-domain test, outward scalar products, and independently enclosed own-policy matrices. Failed inverse, positivity, domain, or final tolerance checks remain failures to certify.

S.5.0.0.2. *Certified current-action responses.* In this convex capital economy the optimal current-action objective is strongly convex with quadratic modulus at least r_0 . If a policy has a full recursive-cost gap at most E at a given initial state, its current action differs from the optimal action by at most $\sqrt{E/r_0}$. The mean of its d action coordinates therefore differs by at most $\sqrt{E/(dr_0)}$. Pairing two such intervals gives a valid current-action response band under a change in risk preferences with all

future policies reoptimized. No independence or population sampling assertion is needed.

S.6. PROOFS FOR PRIMITIVE-BUDGET RECURSIVE LEARNING

S.6.1. *The Gaussian continuation map*

For symmetric P in the norm domain $2\theta\sigma\|P\|_2 < 1$, the map $\Psi(P) = P(I - 2\theta\Sigma P)^{-1}$ is symmetric. This follows from its convergent series, or from $(I - 2\theta P\Sigma)^{-1}P = P(I - 2\theta\Sigma P)^{-1}$; commutation of P and Σ is not required. Differentiation gives

$$D\Psi(P)[E] = (I - 2\theta P\Sigma)^{-1}E(I - 2\theta\Sigma P)^{-1}.$$

On a segment with $\|P\|_2 \leq p$ and $\gamma = 1 - 2\theta\sigma p > 0$, the Neumann bound therefore gives $\|D\Psi(P)[E]\|_2 \leq \gamma^{-2}\|E\|_2$. The corresponding scalar term in the certainty equivalent has derivative

$$Dh_\theta(P)[E] = \text{tr}\{(I - 2\theta\Sigma P)^{-1}\Sigma E\}.$$

For a positive-semidefinite segment, its norm is bounded by $\text{tr}(\Sigma)\|E\|_2/\gamma$. The limits at $\theta = 0$ are $D\Psi(P)[E] = E$ and $Dh_0(P)[E] = \text{tr}(\Sigma E)$. These distinct powers of γ explain the separate coefficient and constant weights in the policy certificate.

S.6.2. *Proof of Proposition 9*

The assertion is exact at T . Suppose it holds at $t + 1$. Both the proposed center and the unknown exact coefficient lie in the norm ball of radius p_{t+1} about zero. The positive margin guarantees that the exact Gaussian moment exists. The derivative bound on the segment joining the two matrices implies

$$\|\Psi(P_{t+1}) - \Psi(C_{t+1})\|_2 \leq e_{t+1}/\gamma_t^2.$$

Subtract the proposed Bellman equation from exact own-policy evaluation. The triangle inequality gives (38). Positive definiteness of the exact coefficient follows inductively from the positive stage cost and the positive-semidefinite transformed

continuation. The constants in the own-policy values are finite because every Gaussian domain margin is positive. No positivity of the numerical proposal itself is needed.

Writing the exact action residual as $D_t = R_t K_t - \beta B'_t \Psi(P_{t+1}) F_t$ and adding and subtracting the centered expression proves (39), using $\|B'EF\|_F \leq \|B\|_2 \|E\|_2 \|F\|_F$. All inequalities concern exact objects enclosed by the reported arithmetic; the nominal inverse or nominal residual is never used without its enclosure.

S.6.3. A trainable-factor contraction

Let $mI \preceq M \preceq LI$, $W_0 = \sqrt{c}O$ with O orthogonal, $0 < c \leq m$, and $0 < \alpha \leq 1/(4L)$. Set $X_j = W'_j W_j$. The iteration in (43) implies

$$X_{j+1} = (I + \alpha(M - X_j))X_j(I + \alpha(M - X_j)).$$

Since $X_0 = cI$, every X_j commutes with M . In their common eigenbasis, for an eigenvalue λ of M ,

$$x_{j+1} = x_j \{1 + \alpha(\lambda - x_j)\}^2.$$

If $c \leq x_j \leq \lambda$, the update is no smaller than x_j and no larger than λ . To prove the latter statement, factor

$$\sqrt{\lambda} - \sqrt{x_j} \{1 + \alpha(\lambda - x_j)\} = (\sqrt{\lambda} - \sqrt{x_j}) \{1 - \alpha\sqrt{x_j}(\sqrt{\lambda} + \sqrt{x_j})\} \geq 0.$$

The last factor is at least $1 - 2\alpha L > 0$. Subtracting the scalar update also yields

$$0 \leq \lambda - x_{j+1} = (\lambda - x_j) \{1 - 2\alpha x_j - \alpha^2 x_j(\lambda - x_j)\} \leq (1 - \alpha c)(\lambda - x_j).$$

Consequently $cI \preceq X_j \preceq M$ and $\|X_j - M\|_F \leq (1 - \alpha c)^j \sqrt{d} L$. This proves the sufficient cap in (46). The proof trains all factor entries; simultaneous diagonalization is an argument about the iterates, not an eigenvalue oracle supplied to the algorithm.

S.6.4. Proof of Theorem 10

We first prove the primitive envelope and target bounds jointly, backward in time. At T , the exact coefficient is Q_f . Suppose $0 \preceq P_{t+1} \preceq \bar{p}_{t+1}I$ and, at nonterminal

dates, $P_{t+1} \succeq Q_{t+1}$. The risk-domain test then implies

$$dq_{t+1}I \preceq d\Psi(P_{t+1}) \preceq d\bar{p}_{t+1}I/\bar{\gamma}_t$$

for every fitted date. The contraction just proved gives a positive semidefinite fitted matrix $\hat{P} = W'W/d \preceq \Psi(P_{t+1})$ with $\|\hat{P} - \Psi(P_{t+1})\|_F \leq a_t/d$. The fitted actor satisfies

$$K_t = (R_t + \beta B_t' \hat{P} B_t)^{-1} \beta B_t' \hat{P} A_t, \quad \|A_t - B_t K_t\|_2 \leq f_t.$$

Its exact residual therefore obeys

$$\|D_t\|_F = \|\beta B_t'(\hat{P} - \Psi(P_{t+1}))F_t\|_F \leq \beta b_t f_t a_t / d \leq \sqrt{sr_t q_t}.$$

At the terminal actor this residual is zero. Completing the square in the true one-step objective shows that the fitted actor's loss against its true greedy improvement is bounded by $sq_t \|x\|^2$. Comparing that minimum with the fixed reference action $-L_t x$ gives the matrix inequality

$$P_t \preceq Q_t + L_t' R_t L_t + \beta (A_t - B_t L_t)' \Psi(P_{t+1}) (A_t - B_t L_t) + sq_t I \preceq \bar{p}_t I.$$

Here $sq_t I \preceq \eta Q_t$. The lower bound $P_t \succeq Q_t$ follows from own-policy evaluation. This closes the induction without assuming a bound on a target not yet generated.

It remains to bound the recursive policy loss. Apply the lower-subsolution recursion (36). Suppose inductively that $\delta_{t+1} \leq s\lambda_{t+1} \leq q_*$ and $\kappa_{t+1} \leq s\mu_{t+1}$. Its continuation margin is at least $\bar{\gamma}_t$, its full-action residual is at most $\sqrt{sr_t q_t} \leq \sqrt{\eta r_t q_t}$, and its lower-greedy closed-loop bound is at most $\tilde{f}_t$. Thus

$$\delta_t \leq sq_t \mathbf{1}_{\{t < T-1\}} + \frac{\beta \tilde{f}_t^2}{\bar{\gamma}_t^2} s\lambda_{t+1} = s\lambda_t, \quad \kappa_t \leq s\mu_t.$$

The condition $s\lambda_t \leq q_*$ ensures positivity of every required lower continuation because $P_t \succeq Q_t \succeq q_t I$. The terminal allowance is zero, so the induction starts. The full adapted policy comparison in Theorem 8, followed by (42), proves (45).

S.6.5. Numerical acceptance and work

The finite cap bounds ideal hidden-weight updates, not processor time. The implementation may use a smaller step, an enlarged cap, and periodically spaced residual checks. A failed numerical training threshold or a failed risk-domain test remains a failed attempt. For a returned rounded gain, the outward residual-centered evaluation and the separately bounded action error are the acceptance criteria. The experiment charges primitive-envelope construction, own-policy transforms, all factor updates, actor solves, residual checks, successful and failed verification, and durable output. A structural comparator receives the same analytic Gaussian primitives and is timed and verified separately. Access to those primitives is not evidence of a sampling advantage in a model whose continuation must be simulated.

S.7. FROM TRAINING GEOMETRY TO POLICY ACCURACY

The constructive schedule above controls a particular initialization. A returned factor need not have been trained from that initialization. We therefore give a second, a posteriori route which uses both first- and second-order information. It rules out a singular stationary factor without assuming that its rank has already been certified. Throughout this section $W \in \mathbb{R}^{d \times d}$ is the trainable hidden-weight matrix, not a fixed dictionary.

For a symmetric target $M \succeq mI \succ 0$, put

$$f_M(W) = \frac{1}{4} \|W'W - M\|_F^2, \quad G_M(W) = W(W'W - M).$$

The Hessian is a self-adjoint operator on matrices equipped with the Frobenius inner product. Its lower bound below must hold in every matrix direction; a finite selection of curvature probes does not establish that premise.

THEOREM S.6—Gradient–curvature certificate: *Suppose $\nabla^2 f_M(W) \succeq -\kappa I$, where $0 \leq \kappa < m$. Then*

$$\sigma_{\min}(W)^2 \geq \frac{m - \kappa}{3}, \quad \|W'W - M\|_F \leq \sqrt{\frac{3}{m - \kappa}} \|G_M(W)\|_F. \quad (\text{S.15})$$

In particular, a zero-gradient factor satisfying this curvature condition represents the target exactly.

PROOF: For $E = W'W - M$ and an arbitrary direction Z , differentiation gives

$$D^2 f_M(W)[Z, Z] = \frac{1}{2} \|W'Z + Z'W\|_F^2 + \text{tr}(EZ'Z).$$

Let $s = \sigma_{\min}(W)$ and choose a corresponding pair of unit singular vectors u, v , so that $Wv = su$ and $W'u = sv$. These relations also hold at $s = 0$, with left and right null vectors. Set $Z = uv'$. Then $\|Z\|_F = 1$ and

$$-\kappa \leq D^2 f_M(W)[Z, Z] = 3s^2 - v'Mv \leq 3s^2 - m.$$

This proves the lower singular-value bound. Since $m > \kappa$, W is invertible, and $E = W^{-1}G_M(W)$. The Frobenius norm inequality proves the second assertion. *Q.E.D.*

At $W = 0$, the gradient vanishes for every target, although its approximation error is generally large. The Hessian has a direction with curvature at most $-m$. Thus the theorem excludes exactly the false implication from a small gradient alone that would arise at this stationary point. It does not assert that arbitrary nonconvex training attains the required second-order condition. The finite construction in Theorem 10 and the certificate here are distinct results.

S.7.1. An inexact own-policy target

The policy evaluator may supply a symmetric approximation $\tilde{M}$ rather than the exact target. Suppose an independent evaluation account establishes $\|M - \tilde{M}\|_2 \leq e$. Let

$$\|G_{\tilde{M}}(W)\|_F \leq g, \quad \nabla^2 f_{\tilde{M}}(W) \succeq -\tilde{\kappa}I, \quad \|W\|_F \leq w.$$

COROLLARY S.7—Evaluation-aware factor error: If $M \succeq mI$ and $\tilde{\kappa} + e < m$, then

$$\|W'W - M\|_F \leq (g + we) \sqrt{\frac{3}{m - \tilde{\kappa} - e}}. \quad (\text{S.16})$$

A sufficient check for the target floor is $\tilde{M} \succeq (m + e)I$.

PROOF: The gradient difference is $-W(M - \tilde{M})$, with Frobenius norm at most $\|W\|_F e$. The difference of the Hessian operators maps Z to $-Z(M - \tilde{M})$ and has

operator norm at most e . Hence the true Hessian is bounded below by $-(\tilde{\kappa} + e)I$ and the true gradient norm by $g + we$. Apply Theorem S.6. The sufficient target floor follows from the spectral error bound. Notice that a spectral error for the target multiplies the *Frobenius* norm of W here; replacing it by its spectral norm without a dimension allowance would be invalid. *Q.E.D.*

For the recursive economy the exact target at date t is $M_t = d\Psi_\theta(P_{t+1}^\pi)$, evaluated under the returned policy's own future. Proposition 9 supplies the sufficient radius $e_t^M = de_{t+1}/\gamma_t^2$ around $\tilde{M}_t = d\Psi_\theta(C_{t+1})$; the approximation of that transformed center, when numerical, contributes its own additional radius. Neither a target inherited from a different policy nor a nominal matrix inverse supplies this radius without its separate evaluation bound.

Write a_t for the right side of (S.16) divided by d , $\hat{P}_t = W_t'W_t/d$, and let

$$Z_t = R_tK_t - \beta B_t'\hat{P}_t(A_t - B_tK_t)$$

be the actor residual against the fitted continuation. The true full-action residual then satisfies

$$\|D_t\|_F \leq \|Z_t\|_F + \beta\|B_t\|_2\|A_t - B_tK_t\|_2a_t. \quad (\text{S.17})$$

Indeed, subtracting the two residuals leaves $\beta B_t'(\hat{P}_t - \Psi_\theta(P_{t+1}^\pi))(A_t - B_tK_t)$. This identity proves the bound. Substitution in Theorem 8 gives a full adapted-policy certificate whenever its domain, positivity, and implementation conditions also hold. The curvature check is consequently a training diagnostic with an explicit route to economic accuracy, not a replacement for the independent economic stopping rule.

S.7.2. The terminal boundary and the iteration account

The terminal coefficient is known, not learned. Under recursive risk preferences, the last actor must nevertheless transform that known coefficient:

$$K_{T-1} = \{R_{T-1} + \beta B_{T-1}'\Psi_\theta(Q_f)B_{T-1}\}^{-1}\beta B_{T-1}'\Psi_\theta(Q_f)A_{T-1}. \quad (\text{S.18})$$

This is the meaning of direct terminal evaluation. Substituting Q_f for $\Psi_\theta(Q_f)$ in this formula is justified at $\theta = 0$, or when the transform actually equals Q_f , but

not in general. The scalar constant $h_\theta(Q_f)$ does not change this actor. The frozen implementation already uses (S.18); the current manuscript makes the boundary explicit.

The isotropic factor construction also admits the sharper contraction

$$\begin{aligned} \|W_j'W_j - M\|_F &\leq (1 - 2\alpha c)^j \sqrt{d} L, \\ 0 < c &\leq \lambda_{\min}(M), \quad 0 < \alpha \leq (4L)^{-1}, \quad \|M\|_2 \leq L. \end{aligned}$$

To see this, use the common eigenbasis in the constructive proof. With $c \leq x_j \leq \lambda$,

$$\lambda - x_{j+1} = (\lambda - x_j) \{1 - 2\alpha x_j - \alpha^2 x_j (\lambda - x_j)\} \leq (1 - 2\alpha c)(\lambda - x_j).$$

Nonnegativity follows from the previously proved upper bound $x_{j+1} \leq \lambda$, and $1 - 2\alpha c \geq 1/2$. This sharper account justifies a cap based on $-\log(1 - 2\alpha c)$ as well as the looser published cap. It changes neither the frozen experimental budgets nor their interpretation in rounded arithmetic.

For small matrices the replication code represents stored binary inputs as exact rational numbers and checks the whole Hessian by a rational positive-semidefiniteness elimination. Its certificates are conditional on the declared target-radius bound, which must come from evaluation rather than from this algebraic check. This implementation validates the new implication and its failure cases. It is not offered as a scalable replacement for the fifty-dimensional residual verifier, and its timing is not a primary economic comparison.

S.8. CERTIFIED WARM STARTS UNDER CHANGING CONTINUATIONS

A continuation fitted to one economic future can be useful for another without being accurate enough to reuse unchanged. The relevant distinction is between an admissible initial factor and an admissible final policy. This section gives a finite training account for the former and retains the independent economic certificate for the latter. In contrast to the isotropic construction in Section 8, the starting Gram matrix need not commute with the new target. The result also allows explicitly bounded errors in each hidden-weight update.

Let M be a symmetric continuation target satisfying $mI \preceq M \preceq LI$, where $0 < m \leq L$. The target is fixed during one fit, after the candidate's future policy has been finalized. A factor $W \in \mathbb{R}^{p \times d}$, with $p \geq d$, represents the scalar continuation $\|Wx\|^2/d$; every entry of W is trainable. Write $H_j = W_j'W_j$ and consider the implemented update

$$W_{j+1} = W_j\{I - \alpha(H_j - M)\} + \Delta_j, \quad \|\Delta_j\|_F \leq \nu. \quad (\text{S.19})$$

The perturbation account must cover the whole computed step, including products and subtraction, not only the final storage conversion. The theorem is a deterministic implication for all errors satisfying this account.

THEOREM S.8—Perturbation-stable warm-start training: Choose $0 < r \leq m/2$ and $0 < \alpha \leq \{2(L + r)\}^{-1}$, and set

$$\begin{aligned} b &= L + r, & q &= 1 - 3\alpha m/4, \\ z &= 2\sqrt{b}(1 + \alpha r)\nu + \nu^2, & f &= z/(1 - q). \end{aligned} \quad (\text{S.20})$$

Suppose $\|H_0 - M\|_F \leq e_0 \leq r$ and $f \leq r$. Then every iterate in (S.19) satisfies

$$\|H_j - M\|_F \leq f + q^j(e_0 - f) \leq r. \quad (\text{S.21})$$

In particular, if $f < a < e_0$, it suffices to take

$$k = \left\lceil \frac{\log\{(e_0 - f)/(a - f)\}}{-\log q} \right\rceil \quad (\text{S.22})$$

updates to achieve Gram error at most a . If $e_0 \leq a$, no update is required. No simultaneous diagonalization of H_0 and M is assumed.

The proof in Section S.9 uses a Frobenius-norm contraction for a Sylvester operator, rather than the common eigenbasis used by the cold construction. It therefore covers rotated and overparameterized factors and warm starts whose Gram matrices lie on either side of the target. The positive noise floor is economically relevant: a requested continuation tolerance at or below that floor is not certified by the finite-cap formula. More accurate arithmetic or a different verified construction is then required; increasing the iteration count alone does not justify acceptance.

S.8.1. Transport and own-policy evaluation

Suppose the old factor is accompanied by $\|W'_0W_0 - M_{\text{old}}\|_F \leq e_{\text{old}}$ and the newly evaluated target by $\|M - M_{\text{old}}\|_F \leq h$. The computable gate

$$e_{\text{old}} + h \leq r \quad (\text{S.23})$$

certifies an admissible warm start with $e_0 = e_{\text{old}} + h$. A direct enclosure of $\|W'_0W_0 - M\|_F$ is an alternative, potentially sharper gate. Neither gate certifies the old action as accurate in the changed economy. If the gate fails, the controlled cold construction remains available; the failed gate and all subsequent work are charged. A target obtained from the previous regime's policy cannot replace the new policy's own continuation without a separate transport bound.

A numerical evaluator may return a symmetric center $\tilde{M}$ and an independently verified spectral radius e^M for the exact own-policy target M^π . Apply Theorem S.8 to $\tilde{M}$, using certified bounds on that center. If its final Gram error is at most a , then

$$\|W'W - M^\pi\|_2 \leq a + e^M. \quad (\text{S.24})$$

In the recursive capital economy, $M_t^\pi = d\Psi_\theta(P_{t+1}^\pi)$. The residual-centered evaluation in Proposition 9 supplies $e_t^M = de_{t+1}/\gamma_t^2$, with an additional allowance for evaluating the transformed center when needed. This is a statement about the returned policy, not the policy used to produce an old warm start.

For a returned actor K_t , put $F_t = A_t - B_tK_t$, $\hat{P}_t = W'_tW_t/d$, and $Z_t = R_tK_t - \beta B'_t\hat{P}_tF_t$. Its exact full-action residual obeys

$$\|D_t\|_F \leq \|Z_t\|_F + \frac{\beta\|B_t\|_2\|F_t\|_F}{d}(a_t + e_t^M). \quad (\text{S.25})$$

The Frobenius norm of F_t , rather than its spectral norm without a dimension allowance, is essential when e_t^M is only a spectral radius. Substituting these residuals in Theorem 8, with its moment, positivity, and implementation conditions, gives the same full adapted-policy comparison as before. The training theorem changes the construction of the candidate, not the economic target or the acceptance rule.

S.8.2. *Reliability and complete work*

The reliability statement is uniform over the class of initial factors and update perturbations just specified. It is not a confidence interval inferred from two or three training seeds. A randomized implementation supported entirely in this verified class inherits the deterministic cap. Whether a rounded evaluator, update routine, and returned policy satisfy the premises is checked separately. Failed domain, transport, precision, or final-policy checks remain failures to certify.

For a sequence of changed futures, the warm-start service account is

$$C_{\text{warm}} = C_{\text{initial}} + \sum_{\ell} \{C_{\text{evaluate},\ell} + C_{\text{gate},\ell} + C_{\text{fit},\ell} + C_{\text{actor},\ell} + C_{\text{verify},\ell} + C_{\text{write},\ell}\}.$$

(S.26)

Failed gates, cold fallbacks, and failed final checks enter this same sum. The cap gives an upper bound on the number of factor updates in an admissible warm fit. Comparing two upper bounds on training iterations does not establish a lower realized cost than a competitor that stops early. An economic work advantage requires complete measured costs, or an upper bound for the warm procedure below an independently justified lower bound for its comparator. In particular, the analytic structural solution in the quadratic economy remains a legitimate comparator. This distinction preserves the adverse frontiers of the earlier experiments while giving a positive, checkable criterion for when an old neural continuation can be retrained safely.

The additional validation exercises below concern this training implication, including noncommuting starts, rectangular factors, bounded perturbations, transport failure, and unattainable precision requests. They are algebraic and implementation checks, not additional independent economic experiments and not a replacement for the previously frozen services.

S.9. PROOFS FOR CERTIFIED WARM STARTS

S.9.1. A noncommuting Gram contraction

Let $E = H - M$ and suppose $\|E\|_F \leq r$. Then $(m - r)I \preceq H \preceq (L + r)I = bI$. The ideal, unperturbed factor step has Gram error

$$E^+ = E - \alpha(EH + HE) + \alpha^2 EHE. \quad (\text{S.27})$$

Diagonalize H , solely for this argument, with eigenvalues $h_i \in [m - r, b]$. The linear map $E \mapsto E - \alpha(EH + HE)$ multiplies each matrix entry in that basis by $1 - \alpha(h_i + h_j)$. These multipliers lie between zero and $1 - 2\alpha(m - r)$ because $2\alpha b \leq 1$. Frobenius invariance and $\|EHE\|_F \leq br\|E\|_F$ give

$$\|E^+\|_F \leq \{1 - 2\alpha(m - r) + \alpha^2 br\} \|E\|_F.$$

Since $\alpha b \leq 1/2$ and $r \leq m/2$,

$$2(m - r) - \alpha br \geq 2m - \frac{5}{2}r \geq \frac{3}{4}m.$$

Thus $\|E^+\|_F \leq q\|E\|_F$, with q as in (S.20). Notice that $0 < q < 1$ and that the diagonalizing basis is that of the current H , not a common basis for H and M . This is why no commutation assumption has entered the proof.

S.9.2. Perturbations and the finite cap

Write $U = W(I - \alpha E)$. Since $\|W\|_2^2 = \|H\|_2 \leq b$, $\|U\|_2 \leq \sqrt{b}(1 + \alpha r)$. Adding Δ to the factor changes the Gram matrix by

$$U'\Delta + \Delta'U + \Delta'\Delta.$$

Its Frobenius norm is at most $2\|U\|_2\|\Delta\|_F + \|\Delta\|_F^2 \leq z$. This estimate is valid for every rectangular $p \times d$ factor. Consequently, while the radius condition holds, $\|H_{j+1} - M\|_F \leq q\|H_j - M\|_F + z$. The scalar recursion with initial value e_0 has the exact solution $f + q^j(e_0 - f)$. Each value is a convex combination of e_0 and f , both at most r . Induction therefore proves that the radius condition continues to hold and establishes (S.21). If $f < a < e_0$, solving the scalar inequality for j yields (S.22). If $e_0 \leq a$, the initial factor already meets the requested Gram threshold.

S.9.3. *Transport and economic residuals*

The triangle inequality proves (S.23). For an inexact target, $\|W'W - M^\pi\|_2 \leq \|W'W - \tilde{M}\|_F + \|\tilde{M} - M^\pi\|_2$, proving (S.24). Finally,

$$D_t - Z_t = \beta B'_t \{\hat{P}_t - \Psi_\theta(P_{t+1}^\pi)\} F_t.$$

The inequality $\|B'EF\|_F \leq \|B\|_2 \|E\|_2 \|F\|_F$ proves (S.25). All state and action quantifiers, Gaussian moment margins, lower-subsolution positivity checks, and implementation allowances are those of Theorem 8; none is inferred from the training gate. In particular, a residual against an old regime's continuation must first pay its own-policy evaluation discrepancy.

S.9.4. *Numerical interpretation of the error account*

For a routine that computes the ideal matrix expression exactly and then rounds each entry to a grid of spacing h , nearest rounding gives $v \leq h\sqrt{pd}/2$. A floating-point implementation that also rounds matrix products must add enclosures for those operations before invoking this bound. The exact-rational regression tests use the former, explicitly specified routine. Their success does not validate an unenclosed BLAS roundoff account. Similarly, the stored scalar cap is checked by exact rational comparison rather than trusting a rounded logarithm at a ceiling boundary. This verification concerns the cap arithmetic, not the economic model's unknown evaluation radius.

S.10. PROOFS FOR BUDGETED WARM-START POLICY CONSTRUCTION

S.10.1. *The mixed-norm residual*

Let $E_t = W'_t W_t - \tilde{M}_t$ and $V_t = \tilde{M}_t - dS_t$. Then

$$D_t - Z_t = \frac{\beta}{d} B'_t (E_t + V_t) F_t.$$

The inequalities $\|B'EF\|_F \leq \|B\|_2 \|E\|_F \|F\|_2$ and $\|B'VF\|_F \leq \|B\|_2 \|V\|_2 \|F\|_F$, together with $\|F_t\|_F \leq \sqrt{d}\|F_t\|_2$, prove (48). The spectral discrepancy also gives

$$\|\hat{S}_t - S_t\|_2 \leq (a_t + e_t^M)/d \leq q_*.$$

These statements do not require the center or the Gram error to commute with the true coefficient.

S.10.2. A joint backward induction

The coefficient bound is exact at the terminal date. Suppose $0 \preceq P_{t+1}^\pi \preceq \bar{p}_{t+1}I$. The positive primitive margin implies $0 \preceq S_t \preceq u_t I$, hence $0 \preceq \hat{S}_t \preceq (u_t + q_*)I$. The actor identity can be written

$$(R_t + \beta B_t' \hat{S}_t B_t) K_t = \beta B_t' \hat{S}_t A_t + Z_t.$$

Its left coefficient is bounded below by $r_t I$. Since $z_t \leq \sqrt{sr_t q_t} \leq v_t$, it follows that

$$\|F_t\|_2 \leq \|A_t\|_2 + \frac{b_t}{r_t} \{\beta b_t(u_t + q_*)\|A_t\|_2 + z_t\} \leq f_t^+.$$

Equation (48) and the second local acceptance test therefore yield $\|D_t\|_F \leq \sqrt{sr_t q_t}$.

Completing the square in the exact one-step action objective bounds the returned actor's excess coefficient over its true greedy improvement by $sq_t I$. Comparison of that minimum with the reference feedback L_t gives

$$P_t^\pi \preceq Q_t + L_t' R_t L_t + \beta(A_t - B_t L_t)' S_t (A_t - B_t L_t) + sq_t I \preceq \bar{p}_t I,$$

because $sq_t I \preceq \eta Q_t$. Positive stage costs give $P_t^\pi \succeq Q_t \succeq q_t I$. This closes the coefficient induction. In particular, the target bound used in the next fit is not inferred from a presumed successful policy comparison.

Now apply the subsolution recursion of Theorem 8. If $\delta_{t+1} \leq s\lambda_{t+1}^+ \leq q_*$, its true margin is at least $\bar{\gamma}_t$ and its lower-greedy closed-loop bound is at most g_t .

Consequently

$$\delta_t \leq sq_t + \frac{\beta g_t^2}{\bar{\gamma}_t^2} s\lambda_{t+1}^+ = s\lambda_t^+, \quad \kappa_t \leq \beta \left(s\mu_{t+1}^+ + \frac{\tau s\lambda_{t+1}^+}{\bar{\gamma}_t} \right) = s\mu_t^+.$$

The zero terminal allowances start this induction. The allocation preserves the lower subsolution's required positivity at every noninitial date. The recursive full-policy theorem and the initial-radius inequality then prove (53). Neither state sampling nor a restriction to linear competitors enters this argument.

S.10.3. *Finite arithmetic and fallback*

For rational matrices and a finalized rational feedback, the Gaussian coefficient map uses matrix multiplication and an inverse of $I - 2\theta\Sigma P$. A strict norm-domain margin implies invertibility. The own-policy coefficient and the next fitted actor can therefore be computed by finitely many rational operations. The state-independent log-determinant constant is not an input to these actor equations; its contribution to the policy comparison is already paid by μ_t^+ . The scalar bounds can be enlarged rationally, and squared inequalities can be checked without treating rounded square roots as exact.

With exact evaluation and actor solves, take $e_t^M = z_t = 0$ and select a positive rational a_t satisfying (52). The exact-product, nearest-dyadic update has perturbation norm at most $2^{-b}\sqrt{pd}/2$ at b storage bits. Its Gram noise floor tends to zero as b tends to infinity. For a certified warm basin, choose b so that the floor is below a_t and apply Theorem S.8. Its recurrence is checked with rational upper bounds for the square roots. If the initial error already lies below a_t , no factor update is needed.

On failure of the warm gate, choose rational $c = s_0^2$ with $0 < c \leq m_t$ and $W_0 = s_0 I$ at a nonterminal date. The controlled cold contraction in Section S.6, evaluated without rounding its products, reaches the same positive tolerance in finitely many rational updates. This is an existence and termination statement for a specified arithmetic procedure, not a practical bit-complexity bound for all exact rational implementations. It does not justify discarding failed gates from a measured service cost. The terminal transformed coefficient is supplied directly. These observations prove Corollary 12.

S.11. PROOFS FOR PRECISION-ADAPTIVE REFRESH

PROOF OF THEOREM 13: Write $Y = WM^{-1/2}$ and $C = Y'Y = I + E$. The ideal update in (54), followed by right multiplication by $M^{-1/2}$, is $\bar{Y} = Y(3I - C)/2$. Consequently

$$\bar{Y}'\bar{Y} = C(3I - C)^2/4, \quad \bar{Y}'\bar{Y} - I = (-3E^2 + E^3)/4.$$

These identities hold for rectangular Y ; they require no commutation between $W'W$ and M . The matrices in the right-hand identities commute because they are

polynomials in the single symmetric matrix E . If $\|E\|_2 \leq \rho \leq r$, the second identity gives $\|\bar{Y}'\bar{Y} - I\|_2 \leq c_r \rho^2$. Moreover, for every eigenvalue $e \in [-r, r]$,

$$0 \leq (1 + e)(2 - e)^2/4 = 1 + e^2(e - 3)/4 \leq 1.$$

Hence $\|\bar{Y}\|_2 \leq 1$. Put $Z = \Delta M^{-1/2}$. The implemented Gram matrix differs from the ideal one by $\bar{Y}'Z + Z'\bar{Y} + Z'Z$, of norm at most $2\nu + \nu^2$. The allocated precision therefore implies

$$\|E_{j+1}\|_2 \leq c_r \rho_j^2 + 2\nu_j + \nu_j^2 \leq \rho_j^2 = \rho_{j+1}.$$

Induction proves (56), since $\rho_j^2 \leq \rho_j \leq r$. Solving $\rho_0^{2^K} \leq \tau$ gives (57). The zero-update cases are immediate.

Entrywise nearest-grid rounding has Frobenius error at most $\sqrt{pd}2^{-b-1}$. Multiplication by $M^{-1/2}$ and $\|M^{-1/2}\|_2 \leq m^{-1/2}$ give the specified upper bound ν . In particular, $\nu \leq (1 - c_r)\rho_j^2/3$ is sufficient: it implies $\nu \leq 1$ and thus $2\nu + \nu^2 \leq 3\nu \leq (1 - c_r)\rho_j^2$. An admissible integer therefore always exists, and the first admissible positive integer satisfies

$$b_j \leq C + 2\log_2(1/\rho_j)$$

for a finite constant C depending only on A and r . If $K \geq 1$ is the first successful update count, $\rho_0^{2^{K-1}} > \tau$. Hence

$$\sum_{j < K} b_j \leq CK + 2\log_2(1/\rho_0)(2^K - 1)$$

$$< CK + 4\log_2(1/\tau),$$

which proves (59). A zero-update fit contributes an empty sum. This bound is on stored fractional precision, not on the bit lengths of intermediate rational products.

Finally, the target is fixed throughout the fit. Its exact inverse can be constructed once, checked against M , and applied to each successive Gram matrix. The inverse applications and their residual checks are still performed and charged. No inverse is needed when the initial factor is accepted. This establishes the final accounting assertion without claiming that the training or the checks have zero cost. *Q.E.D.*

PROOF OF THE TARGET-CHANGE BOUND: The old certificate and target comparison give

$$(1 - \rho_{\text{old}})(1 - \omega)M_{\text{new}} \preceq W'W \preceq (1 + \rho_{\text{old}})(1 + \omega)M_{\text{new}}.$$

The upper relative excess is $\rho_{\text{old}} + \omega + \rho_{\text{old}}\omega$; the lower relative deficit is $\rho_{\text{old}} + \omega - \rho_{\text{old}}\omega$, which is no greater. The larger quantity thus certifies the proposed new basin radius. The calculation concerns only the old factor, not an inverse or a policy that has been accepted under a different target. *Q.E.D.*

PROOF OF COROLLARY 14: For the current center $\tilde{M}_t$, the relative Gram enclosure yields

$$\|W'_t W_t - \tilde{M}_t\|_F \leq \sqrt{d} \|\tilde{M}_t\|_2 \|E_t\|_2 \leq \sqrt{d} L_t \tau_t \leq a_t.$$

The target-change gate, or a direct gate, supplies the initial basin. The precision theorem supplies this inequality in finitely many steps for each positive allowance, provided the implementation supplies the required precision. The terminal transformed coefficient is evaluated directly. Working backward after finalizing each future policy therefore supplies exactly the Gram, evaluation, and actor premises of Theorem 11. That theorem proves the coefficient envelope and the all-action Bellman comparison jointly, and gives the displayed initial-state policy bound. No inference from sampled states, a finite action catalogue, or a small fitting loss is used in this implication. An independently bounded discrepancy between the stored policy and its executed actions must be added through its own implementation account. *Q.E.D.*

S.12. COMPLETE FACTORIAL AND TIMING LEDGER

The following entries summarize all 96 quadratic economy–method objects. Each has three isolated repetitions. Times are complete-service seconds; the range gives the observed minimum and maximum. U counts updates, B target-inverse builds and F failed policy checks in the final tuning trial. The machine-readable audit and original records additionally preserve all tuning trials, actor solves, own-policy evaluation dates, moment margins, native precision, process memory and startup-

inclusive clocks. Methods: FR/FC are fixed64 rebuilt/cached; AR/AC are adaptive rebuilt/cached; TC is tuned native cached; S is structural.

Object	Method	Median	Min	Max	U	B	F
tol 0.0001	AC	0.433	0.417	0.433	11	3	15
tol 0.0001	AR	0.421	0.418	0.422	11	11	15
tol 0.0001	FC	0.415	0.415	0.421	11	3	15
tol 0.0001	FR	0.416	0.416	0.420	11	11	15
tol 0.0001	S	0.170	0.168	0.173	0	0	4
tol 0.0001	TC	0.424	0.423	0.428	11	3	15
tol 0.01	AC	0.352	0.350	0.356	8	3	12
tol 0.01	AR	0.351	0.349	0.358	8	8	12
tol 0.01	FC	0.353	0.352	0.354	8	3	12
tol 0.01	FR	0.354	0.352	0.354	8	8	12
tol 0.01	S	0.175	0.172	0.181	0	0	4
tol 0.01	TC	0.356	0.351	0.356	8	3	12
tol 1e-06	AC	0.444	0.443	0.450	12	3	16
tol 1e-06	AR	0.441	0.441	0.449	12	12	16
tol 1e-06	FC	0.442	0.440	0.444	12	3	16
tol 1e-06	FR	0.448	0.440	0.456	12	12	16
tol 1e-06	S	0.171	0.170	0.172	0	0	4
tol 1e-06	TC	0.451	0.438	0.453	12	3	16
tol 1e-08	AC	0.500	0.485	0.503	14	3	18
tol 1e-08	AR	0.489	0.483	0.500	14	14	18
tol 1e-08	FC	0.482	0.481	0.484	14	3	18
tol 1e-08	FR	0.487	0.485	0.495	14	14	18
tol 1e-08	S	0.169	0.169	0.170	0	0	4
tol 1e-08	TC	0.491	0.486	0.492	14	3	18
tol 1e-10	AC	0.486	0.483	0.499	14	3	18
tol 1e-10	AR	0.487	0.486	0.491	14	14	18
tol 1e-10	FC	0.491	0.484	0.509	14	3	18

	Object	Method	Median	Min	Max	U	B	F	
1									1
2									2
3	tol 1e-10	FR	0.491	0.486	0.500	14	14	18	3
4	tol 1e-10	S	0.175	0.173	0.357	0	0	4	4
5	tol 1e-10	TC	0.488	0.488	0.502	14	3	18	5
6	moment	AC	0.490	0.488	0.507	14	3	18	6
7	moment	AR	0.492	0.490	0.501	14	14	18	7
8	moment	FC	0.479	0.476	0.491	13	3	17	8
9	moment	FR	0.474	0.469	0.479	13	13	17	9
10	moment	S	0.171	0.171	0.178	0	0	4	10
11	moment	TC	0.495	0.495	0.507	14	3	18	11
12	grid 16-8-16.0	AC	2.481	2.476	2.533	42	7	50	12
13	grid 16-8-16.0	AR	2.474	2.463	2.636	42	42	50	13
14	grid 16-8-16.0	FC	2.542	2.459	2.543	42	7	50	14
15	grid 16-8-16.0	FR	2.518	2.471	2.568	42	42	50	15
16	grid 16-8-16.0	S	0.476	0.470	0.479	0	0	8	16
17	grid 16-8-16.0	TC	2.493	2.456	2.541	42	7	50	17
18	grid 32-4-16.0	AC	0.859	0.856	0.876	18	3	22	18
19	grid 32-4-16.0	AR	0.855	0.850	0.875	18	18	22	19
20	grid 32-4-16.0	FC	0.905	0.887	0.909	19	3	23	20
21	grid 32-4-16.0	FR	0.892	0.889	0.895	19	19	23	21
22	grid 32-4-16.0	S	0.225	0.224	0.226	0	0	4	22
23	grid 32-4-16.0	TC	0.857	0.855	0.858	18	3	22	23
24	grid 4-12-4.0	AC	3.683	3.648	3.708	44	11	56	24
25	grid 4-12-4.0	AR	3.612	3.553	3.614	44	44	56	25
26	grid 4-12-4.0	FC	3.645	3.556	3.725	44	11	56	26
27	grid 4-12-4.0	FR	3.636	3.533	3.654	44	44	56	27
28	grid 4-12-4.0	S	0.856	0.839	0.861	0	0	12	28
29	grid 4-12-4.0	TC	3.633	3.551	3.752	44	11	56	29
30	grid 8-4-8.0	AC	0.532	0.527	0.549	15	3	19	30
31	grid 8-4-8.0	AR	0.536	0.534	0.540	15	15	19	31
32	grid 8-4-8.0	FC	0.534	0.526	0.538	15	3	19	32

	Object	Method	Median	Min	Max	U	B	F	
1									1
2									2
3	grid 8-4-8.0	FR	0.527	0.526	0.534	15	15	19	3
4	grid 8-4-8.0	S	0.178	0.176	0.180	0	0	4	4
5	grid 8-4-8.0	TC	0.529	0.528	0.537	15	3	19	5
6	change 0.25-cold	AC	0.448	0.446	0.449	12	3	16	6
7	change 0.25-cold	AR	0.450	0.448	0.454	12	12	16	7
8	change 0.25-cold	FC	0.445	0.445	0.445	12	3	16	8
9	change 0.25-cold	FR	0.445	0.441	0.447	12	12	16	9
10	change 0.25-cold	S	0.172	0.168	0.174	0	0	4	10
11	change 0.25-cold	TC	0.446	0.437	0.451	12	3	16	11
12	change 0.25-warm	AC	0.872	0.863	0.880	12	3	16	12
13	change 0.25-warm	AR	0.864	0.858	0.901	12	12	16	13
14	change 0.25-warm	FC	0.863	0.853	0.865	12	3	16	14
15	change 0.25-warm	FR	0.882	0.876	0.919	12	12	16	15
16	change 0.25-warm	S	0.598	0.584	0.602	0	0	4	16
17	change 0.25-warm	TC	0.872	0.869	0.884	12	3	16	17
18	change 1.005-cold	AC	0.439	0.437	0.442	12	3	16	18
19	change 1.005-cold	AR	0.448	0.442	0.473	12	12	16	19
20	change 1.005-cold	FC	0.441	0.436	0.460	12	3	16	20
21	change 1.005-cold	FR	0.440	0.439	0.473	12	12	16	21
22	change 1.005-cold	S	0.173	0.169	0.173	0	0	4	22
23	change 1.005-cold	TC	0.447	0.441	0.464	12	3	16	23
24	change 1.005-warm	AC	0.679	0.658	0.680	3	3	7	24
25	change 1.005-warm	AR	0.682	0.668	0.691	3	3	7	25
26	change 1.005-warm	FC	0.659	0.657	0.677	3	3	7	26
27	change 1.005-warm	FR	0.665	0.662	0.689	3	3	7	27
28	change 1.005-warm	S	0.605	0.602	0.622	0	0	4	28
29	change 1.005-warm	TC	0.666	0.660	0.667	3	3	7	29
30	change 4.0-cold	AC	0.451	0.440	0.452	12	3	16	30
31	change 4.0-cold	AR	0.444	0.443	0.452	12	12	16	31
32	change 4.0-cold	FC	0.446	0.436	0.456	12	3	16	32

Object	Method	Median	Min	Max	U	B	F
change 4.0-cold	FR	0.441	0.440	0.448	12	12	16
change 4.0-cold	S	0.172	0.171	0.173	0	0	4
change 4.0-cold	TC	0.450	0.437	0.462	12	3	16
change 4.0-warm	AC	0.859	0.856	0.882	12	3	16
change 4.0-warm	AR	0.873	0.866	0.876	12	12	16
change 4.0-warm	FC	0.872	0.863	0.944	12	3	16
change 4.0-warm	FR	0.871	0.870	0.906	12	12	16
change 4.0-warm	S	0.594	0.588	0.604	0	0	4
change 4.0-warm	TC	0.870	0.869	0.890	12	3	16

S.12.1. *Nonlinear timing dispersion*

Object	Median	Min	Max	Peak RSS (KiB)
nonlinear-frontier	6.077	5.993	6.086	45700
neural-1.0-0.0	2.846	2.840	2.848	313708
neural-1.0-1.0	5.011	4.984	6.237	314288
neural-4.0-0.0	2.999	2.976	3.005	313744
neural-4.0-1.0	6.222	6.220	6.225	314420
capital-1.0-0.0	0.240	0.240	0.241	45400
capital-1.0-1.0	1.311	1.310	1.325	45400
capital-4.0-0.0	0.409	0.408	0.410	45400
capital-4.0-1.0	2.564	2.562	2.569	45400

The frontier is one complete four-mesh service per repetition. Its internal cumulative construction clocks are distinct from this full-service clock. Peak RSS is process high-water memory, not exclusive algorithm storage.

S.13. ARITHMETIC, DIMENSION, AND THE NEURAL VERIFICATION ACCOUNT

This appendix supplies the arithmetic and work details for Section 12 of the article. The trained functions at both successive dates are inputs to the certificate.

Approximation of those functions by their former training targets is not an allowed operation.

S.13.1. *Exact activation regions and the terminal discrepancy*

Write the stored scalar network as

$$f(x) = b_0 + \sum_{j=1}^m v_j(w_j x + b_j)_+.$$

All coefficients are rational when read as the exact values of their binary representations. Units with $w_j = 0$ contribute constants. For the remaining units, order and deduplicate the rational roots $-b_j/w_j$ in $[0, 1]$. Between consecutive roots the active set is constant, and summing its coefficients gives an exact affine function $A_k x + B_k$. Negative slopes do not require a separate approximation rule. The maximum of $|A_k|$ over nonempty regions is an all-state Lipschitz bound. This maximum can be much smaller than the sum of absolute weights.

PROPOSITION S.9—Evaluation after rational region compilation: *Suppose every rounded root differs from its exact root by at most e_r , while order is retained. Let $L = \max_k |A_k|$. If region selection at x uses the rounded roots, evaluating the selected exact affine extension instead of $f(x)$ makes an error no larger than $2Le_r$. If several rounded roots coincide, the same bound holds provided every displaced boundary between the chosen and true regions lies within e_r of x . Outward coefficient evaluation at a midpoint x_c of an input interval I and an additional $L \text{rad}(I)$ then enclose $f(I)$, after charging midpoint rounding as well.*

PROOF: An incorrectly selected region is separated from the true one only by roots whose rounded comparisons differ at x . Such roots are within e_r of x . Choose the boundary of the selected region closest to x ; call it r . The selected affine function equals f at r by continuity. Both f and that affine function have Lipschitz constant at most L . Their difference at x is therefore at most $2L|x - r| \leq 2Le_r$. For an interval, add the Lipschitz variation from its midpoint and the outward arithmetic enclosure of the selected affine evaluation. *Q.E.D.*

For the terminal primitive, split further at the cost kink $3/8$. On every resulting interval $f_T - g$ is a quadratic polynomial with rational coefficients. A continuous quadratic achieves its extrema at an endpoint or at its stationary point if that point is in the interval. Testing those points gives rational global endpoints. The terminal error is generally nonzero. In the executed study, 66 candidates yield a terminal interval approximately $[-0.000668000, 0.000410383]$ for every economy, because the terminal network and terminal primitive are shared.

S.13.2. *Rounding the specified execution graph*

A range-error pair consists of an exact range $[a, b]$ and a nonnegative rational error e . It means that an ideal node y lies in $[a, b]$ and its implemented value differs by at most e . Put $M([a, b]) = \max(|a|, |b|)$. For separate round-to-nearest binary32 operations, use $u = 2^{-24}$ and $\alpha = 2^{-150}$. If a sum has pre-rounding propagated error $e_s = e_1 + e_2$ and ideal range I_s , its new error is

$$e_s + u\{M(I_s) + e_s\} + \alpha.$$

For a product, first put

$$e_p = M(I_1)e_2 + M(I_2)e_1 + e_1e_2;$$

then add $u\{M(I_1I_2) + e_p\} + \alpha$. These formulae follow by expanding $(y_1 + \delta_1)(y_2 + \delta_2) - y_1y_2$ and applying the elementary rounding inequality. Rational range propagation is performed separately. Conversion of the state, action, price, and shock is included at the input nodes. Clipping to a representable closed interval is nonexpansive and cannot increase this error.

The deposited implementation applies these recurrences to the operation order used in the binary32 transition, cost, and terminal routines. Its verification test compares 1,032 native executions with exact rational execution of the corresponding ideal graphs. Those finite tests detect implementation mistakes; the range induction, not the tests, supplies the all-state guarantee. The maximum observed error is less than 0.224 of the corresponding uniform allowance. No conclusion is drawn for a fused multiply-add implementation or a platform that flushes subnormals to zero.

Nearest-node policy selection deserves a separate argument because the actor is discontinuous. For a true state x and acquired state $\hat{x}$, a node selected nearest to $\hat{x}$ is at most $h_x + |\hat{x} - x|$ from x . The chosen-action and Bellman-minimum comparisons each pay $L_x|\hat{x} - x|$. There is consequently no need for an unjustified Lipschitz bound on the actor itself. At the dyadic grids in this study, stored action values and state cutpoints are exactly representable in binary32; that fact does not remove errors in state acquisition, transition, or cost evaluation.

S.13.3. Continuous innovations and multidimensional verification

The direct certificate is not tied to scalar states or finite-support uncertainty. This extension states explicitly what additional work is required; it does not identify a grid verifier with a dimension-free method. Let $X \subset \mathbb{R}^{d_x}$, $A \subset \mathbb{R}^{d_a}$, and $Z \subset \mathbb{R}^{d_z}$ be compact. The innovation has a known law and admits a deterministic quantizer q with $\|Z - q(Z)\| \leq h_z$ almost surely. The probabilities of the quantizer cells are supplied exactly or by enclosing arithmetic. Take a law-invariant, monotone, cash-invariant one-step risk functional, evaluated conditionally on the current state and action. Expectation and entropic risk satisfy this requirement. Suppose the transition is Lipschitz in the innovation with constant $L_{Fz,t}$. If stage costs also depend on the innovation, add their innovation Lipschitz contribution in the definition below.

THEOREM S.10—A dimension-explicit neural certificate with quantized innovations: *Let the hypotheses of the direct neural certificate hold on X and A . Put $\zeta_t = \beta_t L_{t+1} L_{Fz,t} h_z$. Construct nodal intervals using $q(Z)$ rather than Z . If (ℓ_t, u_t, η_t) are computed from these intervals, then*

$$(\ell_t - \zeta_t, u_t + \zeta_t, \eta_t + 2\zeta_t)$$

are valid residual and actor bounds for the unquantized economy. The global policy certificate increases by at most $4 \sum_{t < T} B_t \zeta_t$.

Let K_x, K_a, m be the numbers of state nodes, action nodes, and quantizer cells. Let P_t count scalar arithmetic operations in a forward network evaluation, $C_{F,t}$ the transition and cost evaluation work, and $C_{\rho,t}(m, p)$ the work to enclose one risk evaluation at precision p .

In units of certified precision- p arithmetic, a streaming direct verification uses

$$W \leq C \sum_{t < T} K_x K_a \{m(C_{F,t} + P_{t+1}) + C_{\rho,t}(m, p)\} + C \sum_{t \leq T} K_x P_t + W_T + W_{i0}.$$

Here W_T certifies the terminal discrepancy and W_{i0} includes serialization and the durable evidence writes. For a batch of b state nodes, its working storage, in scalar words, is at most

$$O\left(\sum_{t \leq T} P_t + TK_x d_a + md_z + bK_a m w_{\max} + K_x\right),$$

where $w_{\max}$ bounds the largest network layer width and $b \leq K_x$. The policy output itself requires $TK_x d_a$ words. Tensor covers of bounded boxes can be chosen with

$$K_x = O(h_x^{-d_x}), \quad K_a = O(h_a^{-d_a}), \quad m = O(h_z^{-d_z}).$$

The hidden constants depend on the domains and the norm, not on training convergence. A finite refinement schedule is charged by summing these bounds over every attempted resolution, not just the accepted resolution.

PROOF: Couple Z and $q(Z)$ on their defining probability space. The two future network values differ almost surely by at most $L_{t+1} L_{Fz,t} h_z$. Monotonicity and cash invariance bound the difference of their risk evaluations by the same number. Law invariance identifies the quantized risk evaluation with its finite-support calculation. Multiplication by β_t gives a uniform Q discrepancy ζ_t . Taking an infimum does not increase a uniform error. A residual therefore pays ζ_t at each endpoint; a chosen-action excess pays $2\zeta_t$. The residual oscillation and actor allowance together pay $4\zeta_t$. Backward induction gives the stated policy bound.

For each of the $TK_x K_a$ state-action pairs the program evaluates m transitions and networks and then one risk aggregate. Current networks are evaluated at the state nodes as well. Streaming over state batches retains only the batch's forward activations, the nodal extrema and stored actors, the innovation support, and the network coefficients. This proves the operation and storage counts. An equally spaced tensor cover proves the node bounds. Exact enumeration of a policy's innovation tree at Q initial queries adds, separately, order $Q \sum_{t=0}^T m^t$ evaluations and

a depth-first stack of order T times the stored state and risk-aggregation data. This exponential query cost is not hidden inside the Bellman verification count. *Q.E.D.*

The notation $C_{\rho,t}(m, p)$ is intentional. A native addition is not an exact exponential, logarithm, or certified inverse. Bit costs require a specified arithmetic implementation and must multiply the appropriate operation counts by its precision-dependent costs. The complete-step matrix result in the article similarly exposes its lower spectral bound and residual enclosure rather than suppressing conditioning in an unspecified constant. Neural Lipschitz bounds, risk-evaluation ranges, terminal verification, and the admissible condition margins must be reported before either count is used numerically.

A refinement reduces verification slack; it cannot force a fixed inaccurate network to satisfy an arbitrary policy target. More precisely, suppose current-value and quantized Q nodal intervals have widths at most $w_{f,t}$ and $w_{q,t}$. Choosing a minimum-upper-endpoint action gives the following useful upper bound on the certificate:

$$E_f \leq \sum_{t < T} B_t \text{osc}(f_t - T_t f_{t+1}) + B_T \text{osc}(f_T - g) \\ + \sum_{t < T} B_t \{ (2L_t + 4L_{x,t})h_x + 3L_{a,t}h_a + 4\zeta_t + 2w_{f,t} + 3w_{q,t} \}.$$

To check the coefficients, the grid minimum differs from the continuous minimum by between zero and $L_{a,t}h_a$. The residual range therefore pays at most $2L_{a,t}h_a + 2w_{f,t} + 2w_{q,t}$, in addition to its state cover. The selected action pays at most one further $L_{a,t}h_a + w_{q,t}$ and its two state movements. Quantization supplies the four allowances already proved. Thus the refinement tends to the network's residual and terminal oscillations, not automatically to zero.

For prescribed verification slack, this theorem specifies its dependence on state, action and innovation dimension, horizon, network size, precision, and the problem's Lipschitz and conditioning data. It does not establish a competitive high-dimensional work frontier. The new executed nonlinear certificate below remains scalar; the higher dimensions in the quadratic study do not change that fact. A coupled nonlinear scaling comparison with strong sparse, projection, or fitted

dynamic-programming baselines remains an empirical question distinct from this extension theorem.

TABLE II: Neural cost minus the newly optimized spline cost

P	θ	Initial state	Lower endpoint	Upper endpoint
1	0	0.125	0.000177945	0.000177946
1	0	0.250	0.000194975	0.000194976
1	0	0.500	0.000213034	0.000213035
1	0	0.750	0.000156082	0.000263132
1	1	0.125	0.000193876	0.000193877
1	1	0.250	0.000187608	0.000187609
1	1	0.500	0.000197012	0.000197013
1	1	0.750	0.000147061	0.000253192
4	0	0.125	0.000203939	0.000203940
4	0	0.250	0.000139974	0.000139975
4	0	0.500	0.000079894	0.000079895
4	0	0.750	-0.000346677	0.000576982
4	1	0.125	0.000153915	0.000153916
4	1	0.250	-0.000249177	0.000545932
4	1	0.500	0.000087748	0.000087749
4	1	0.750	0.000113495	0.000113496

All intervals are computed by subtracting separately enclosed own-policy values; displayed endpoints are rounded outward. Positive differences mean higher neural cost. Every interval is contained in $[-10^{-3}, 10^{-3}]$. This is a descriptive, state-specific cardinal tolerance, not statistical equivalence or a uniform-in-state comparison.

REFERENCES

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